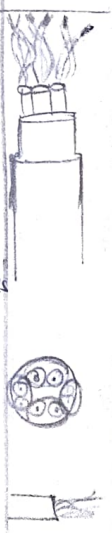
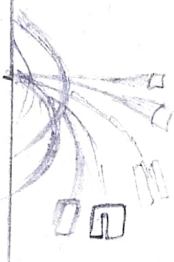
Exp 2 Study of different types of Network cables

Understand different types of network cables

Different types of cables used in network

1) Unshielded Twisted pair
2) Shielded Twisted pair
3) Coaxial cable
4) Fibre Optic cable

Cable type	Category	Max Data Transmission	Adv/ Disadv	Application/ Use	Image
UTP	Category 3	10 bps	**Advantages** • cheaper in cost • Easy to install as they have a smaller overall diameter **Disadvantage** • More prone to EMI and noise	10Base-T Ethernet Fast ethernet Gigabyte ethernet Fast ethernet Giga ethernet	
	Category 5	Upto 100 Mbps			
	Category 5e	1 Gbps			
STP	Category 6, 6a	10 Gbps	**Advantage** • shielded • Faster than UTP • Less susceptible to noise & interference **Disadvantage** • Expensive • Greater installation effort	Gigabit Ethernet, 10G Ethernet (SSM) widely used in datacenters Gigabit Ethernet, 10G Ethernet 100M	
	Category 7	10 Gbps			

Coaxial cable	RG-6 RG-59 RG-11	10-100 Mbps	• high band width • Immune to interference • Low loss bandwidth • Versatile Disadvantages • Limited distance • Cost • Size is bulky	Speed of signal is 500 m. Television network high speed internet connection	
fibre optic cable	Single mode Multi mode	100Gbps	Advantages • high speed • high Security • long distance Disadvantages • Expensive • Required skilled installers	Maximum distance of fibre optic cable is around 100 meters	

(b) Aim:

To make own Ethernet cross over cable / straight cable

Step 1: To start construction of the device, begin by threading shields onto the cable

Step 2: Strip approximately 1.5cm of cable shielded from both ends. The crimping tool has a round area to complete this task

Step 3: After we need to untangle the wires; There should be four "twisted pairs"

Step4: Once the order is correct, bunch them together in a line and if there are any that stick out farther than others Snip them back to create an even level.

step 5: Next push the cable right in. The notch at the end of the plug needs to be just over the shielded and if it isn't, that means that you stripped of too much shielding

step 6: After the wires are securely sitting inside the plug, insert into the crimping tool and push down

step 7: plug it into two system to connect it

Result:

Thus the different types of network cables are studied and connected ethernet cross-over-cable.